

EXPLORATORY DATA ANALYSIS OF YOUTUBE TRENDING VIDEOS DATASET

Content Trends, Viewership
Dynamics & User Engagement



TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

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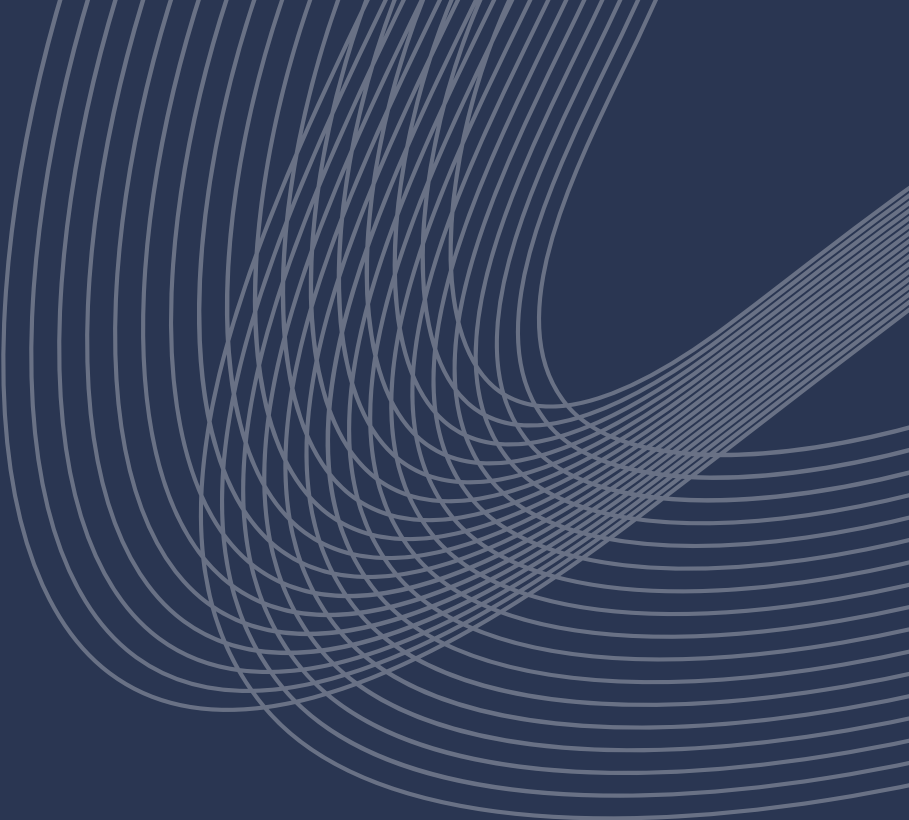


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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

The YouTube platform hosts millions of videos across diverse categories. With massive variations in views, likes, and engagement, it becomes challenging to understand content trends, audience preferences, and what drives video success.

Objective:

To explore the YouTube trending videos dataset and uncover patterns, insights, and trends that help creators and businesses make data driven decisions.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01 Data Collection

- Gathered the Google Play Store dataset from Kaggle

02 Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03 Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04 Exploratory Analysis

- Univariate Analysis: e.g., Ratings, Installs, Prices
- Bivariate Analysis: Rating vs Reviews
- Multivariate Analysis: Reviews vs Rating vs Installs

05 Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED

- Which categories dominate the Play Store, and where is competition highest?
- What is the distribution of Free vs Paid apps across categories and ratings?
- Who are the target audiences of apps (content rating distribution)?
- How do app sizes vary, and what is the optimal size range?
- How do app ratings behave across categories, review volumes, and time?
- What monetization models work best by category and audience?



DATA OVERVIEW

The dataset provides key information about YouTube trending videos, including views, likes, comments, category, publish date, and trending date. Additional details such as channel information and country capture content performance, audience engagement and regional trends.

Data Source: Kaggle

Dataset Size

161470

Records

18

Features

App Diversity

56905

Videos covered

4

Countries

18

Video Categories



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Index	Numerical (Discrete)	Dataset row number
Video_id	String	Unique YouTube video ID
Trending_date	Date	Date the video trended on YouTube
Title	String	Video title
Channel_title	String	Publishing channel name
Category_id	Numerical (Discrete)	YouTube category ID
Publish_date	Date	Video publication date
Time_frame	Categorical	Price of the app (0 indicates a free app)
Published_day_of_week	Date	Day of the week published
Publish_country	Categorical	Country of publication
Tags	String	The tags or keywords associated with the video.
Views	Categorical	The number of views received by a particular video
Likes	Numerical (Discrete)	The number of likes received by a particular video
Dislikes	Numerical (Discrete)	The number of dislikes received by a particular video
Comment_count	Numerical (Discrete)	The number of comments received by a particular video
Comments_disabled	Boolean	Whether comments are disabled
Ratings_disabled	Boolean	Whether ratings are disabled
Video_error_or_removed	Boolean	Whether the video is unavailable, removed, or errored

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- **Columns requiring cleaning or type conversion:** trending_date, publish_date
- **Columns with notable anomalies or invalid entries:**
 - tags (very long strings, |-separated values)
 - publish_country (may require standardization with short forms)



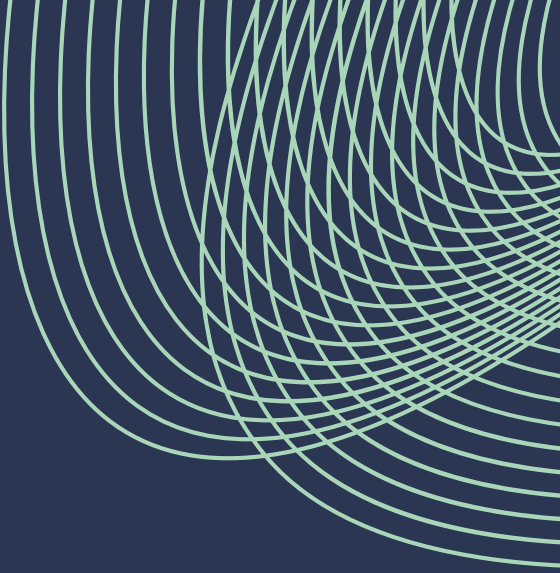
DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

- Features such as **trending_date**, **publish_date** were standardized with respect to data type conversion.
- Short forms the **publish_country** feature were standardized.
- The feature **tags** was transformed into a clean, standardized strings by splitting, normalizing, deduplicating and removing irrelevant tags.
- New features **days_to_trend** & **category_name** were created from **trending_date**, **publish_date** & **category_id** respectively.
- **index** and **video_id** was removed as it is a technical detail with limited business relevance.





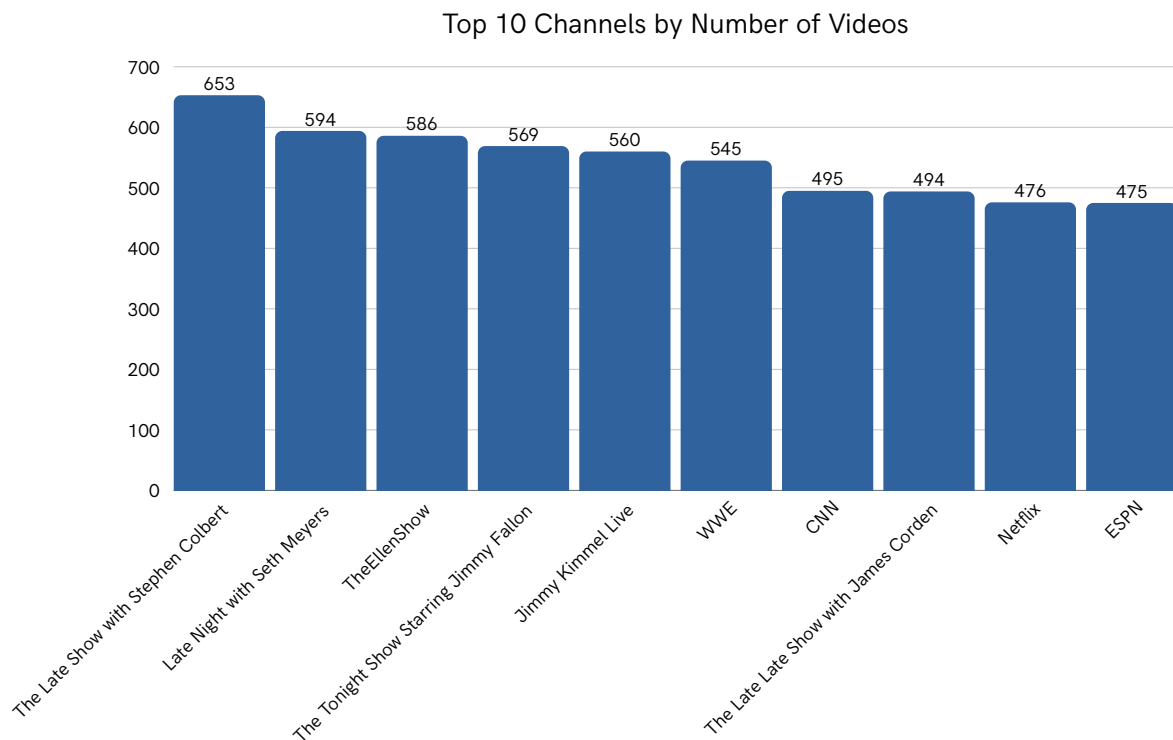
INSIGHTS



Trending Exposure Is Concentrated Among a Few Dominant Channels

5,357

Videos from Top 10 Channels



Key observations

- A small group of channels (late-night shows, major media brands) dominate the number of videos, while most creators appear far less frequently.

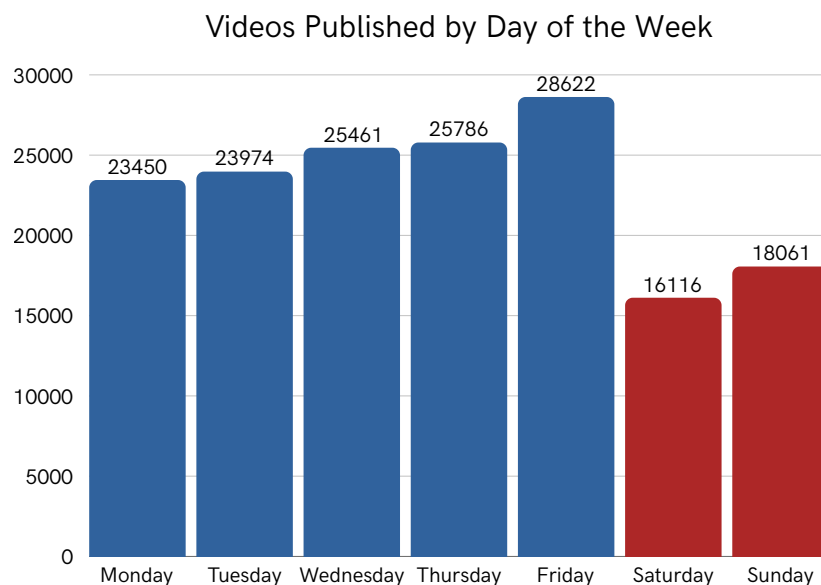
Business Insights

- Visibility is heavily concentrated among established brands.
- New or smaller channels need niche content or viral strategies to compete effectively.

Video Publishing Peaks on Weekdays and Drops on Weekends

23,067

Average Weekly Upload Volume



Key observations

- Video publishing peaks on weekdays, especially Friday and drops significantly over the weekend.

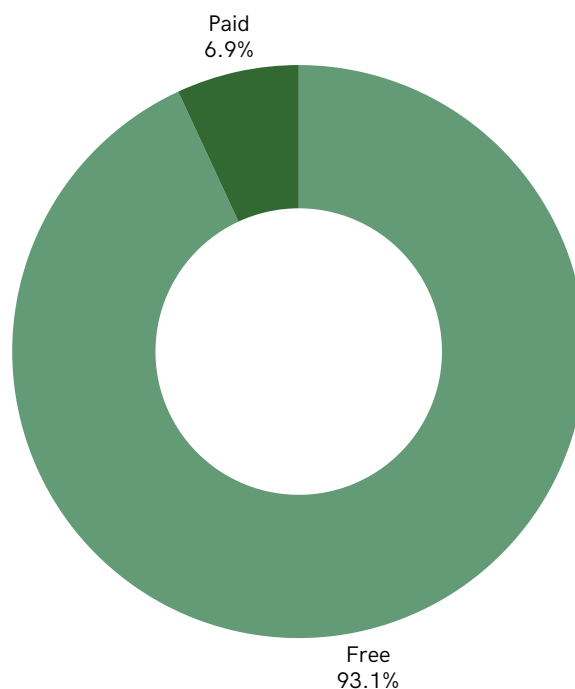
Business Insights

- Uploading during weekdays, particularly **Thursday-Friday** can improve visibility and engagement, while weekend uploads face lower competition but reduced overall activity.

Most apps are free to download

8,719

Apps are free to download



Key observations

- Majority of apps on the Play Store are free (93.1%), with paid apps representing a much smaller share (6.9%).

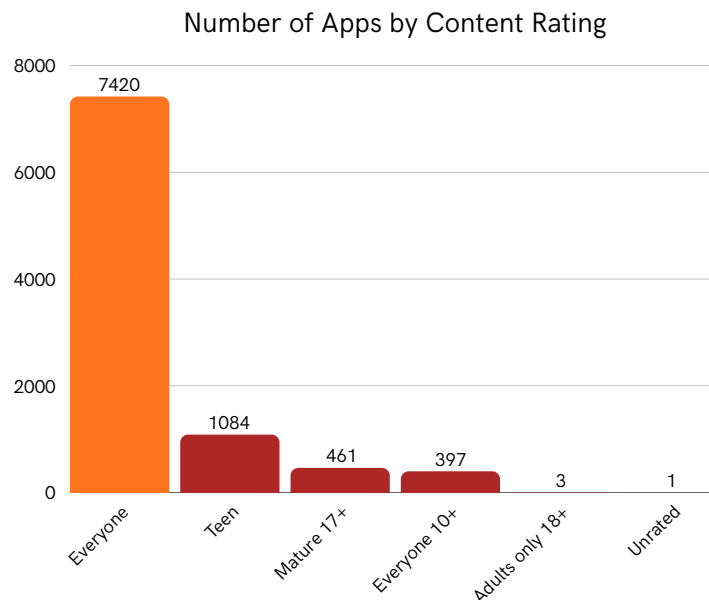
Business Insights

- The dominance of free apps suggests that ad-based or freemium models are more common than upfront paid pricing.

Apps primarily target general audiences

79.2%

Apps targeted general audience



Key observations

- Apps rated for general audiences (Everyone) significantly outnumbered apps targeting teens & mature audiences.

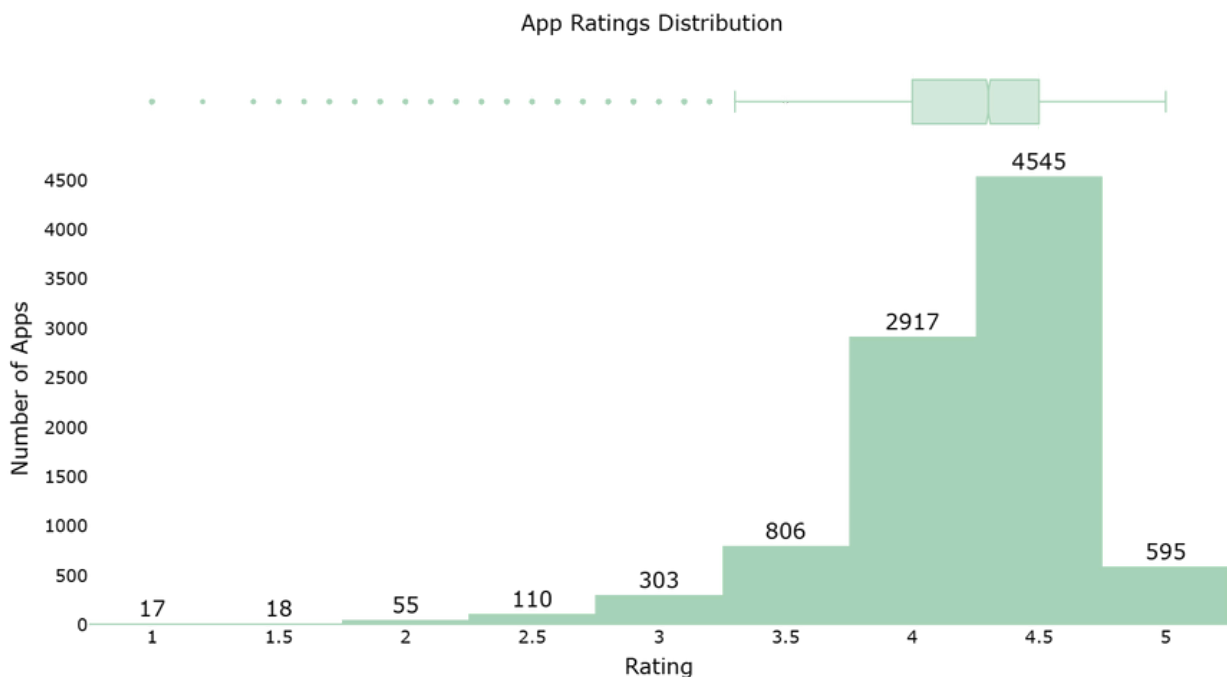
Business Insights

- Developers may favour "Everyone"-rated apps to reach a broader audience, increase downloads, and generate higher ad revenue due to fewer content restrictions.

Most apps have high user ratings

4.3

Typical App Rating (Median)



Key observations

- The rating distribution is left-skewed, with most apps clustered between 3.5 and 5 stars. Few apps have low ratings.
- The median rating is 4.3, slightly higher than the mean of 4.1, reflecting the left skew.

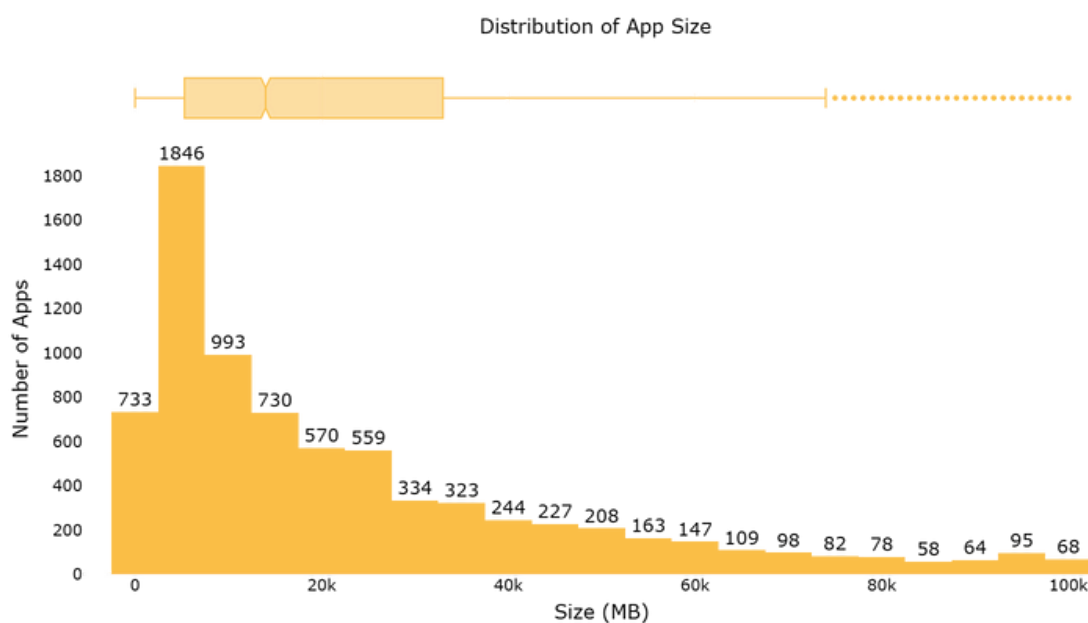
Business Insights

- High average ratings reflect strong competition and high user expectations, while lower-rated apps may struggle to attract or retain users.

Most apps fall within a moderate size range

4.3

Typical App Rating (Median)



Key observations

- Most apps are moderately sized, with few very small or very large ones.
- The median app size is 14 MB, and the mean is 23 MB, indicating that a small number of very large apps pull the average upward.

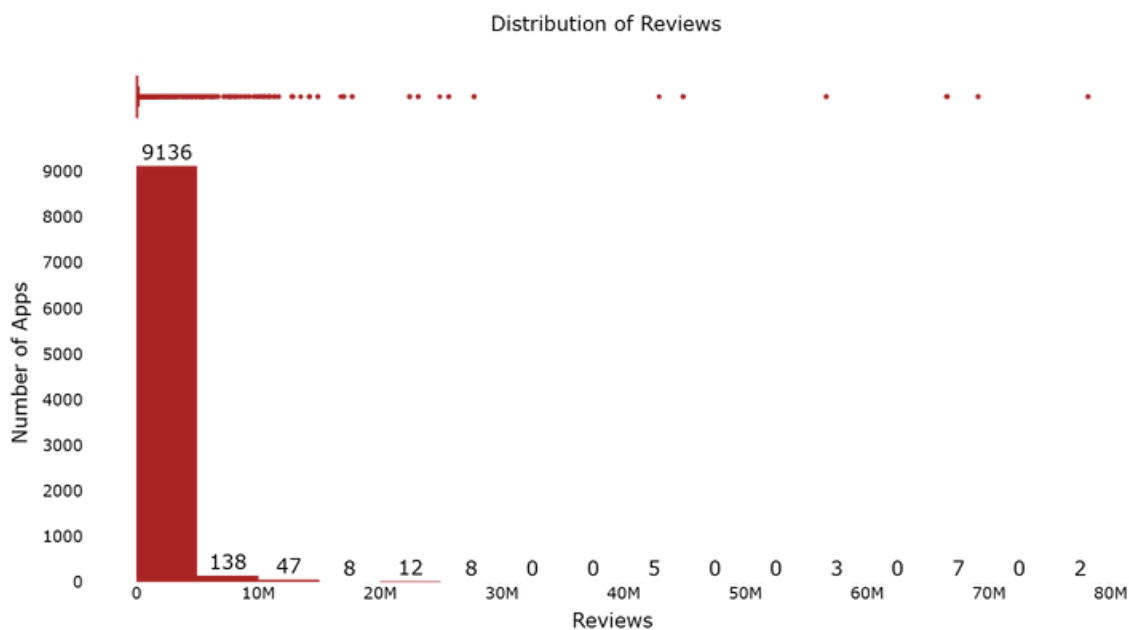
Business Insights

- Developers appear to balance functionality with storage constraints when designing apps, aiming to provide rich features without creating excessively large files.

Popularity Is Concentrated Among Few Apps

5,930

Typical Number of Reviews (Median)



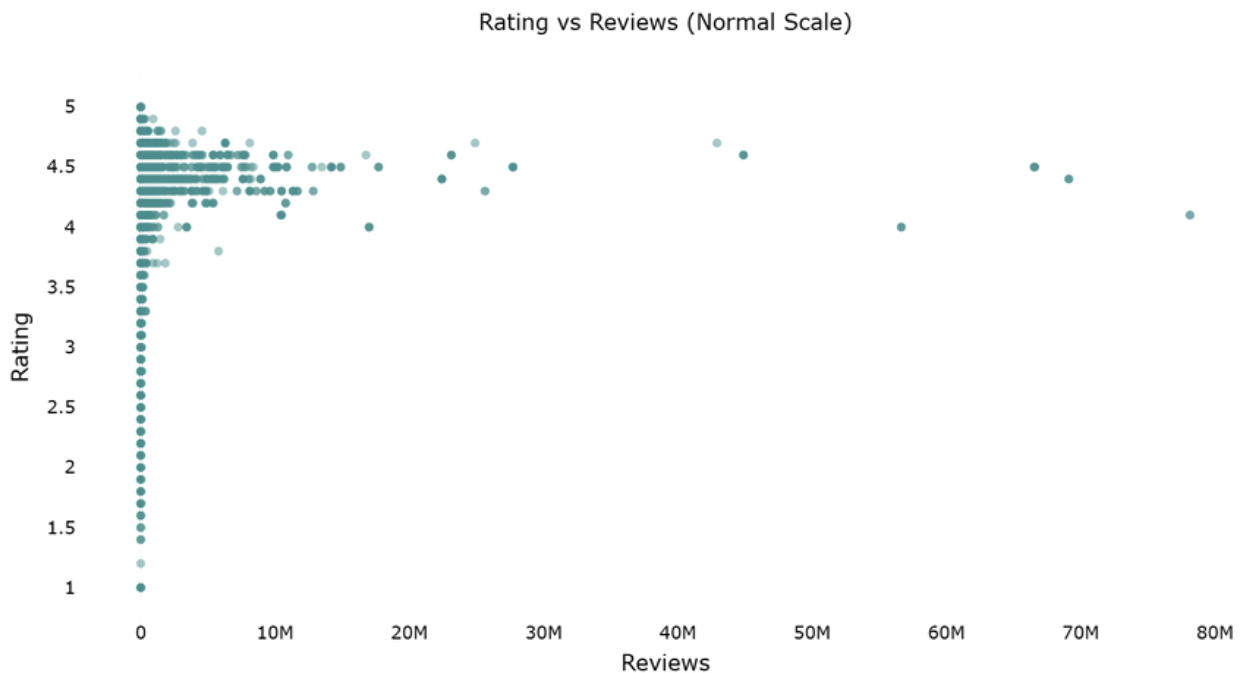
Key observations

- Most apps receive few reviews, while a small number get extremely high counts.
- The median number of app reviews is 5,930, and the mean is 514,050, indicating that a small number of apps with extremely high review counts pull the average upward.

Business Insights

- User engagement is concentrated among a few popular apps, making visibility a key challenge for new apps.

Ratings Stabilize as Review Volume Increases



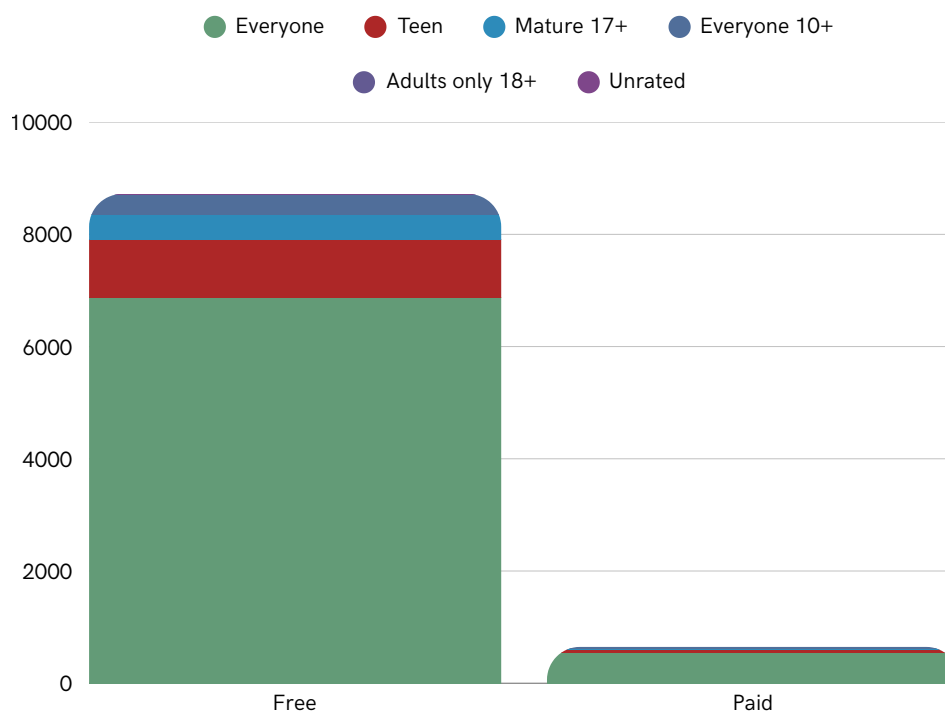
Key observations

- Most apps have few reviews with widely varying ratings, while high-review apps are rare and clustered above 4.
- Rating variability decreases as review count increases, indicating greater stability with more feedback.

Business Insights

- Ratings from low-review apps are noisy; review volume adds credibility and reliability to ratings.
- Encouraging early reviews helps stabilize perceived quality rather than directly boosting ratings.

Upfront Paid Apps Are Uncommon; General Users Access Most Free Apps



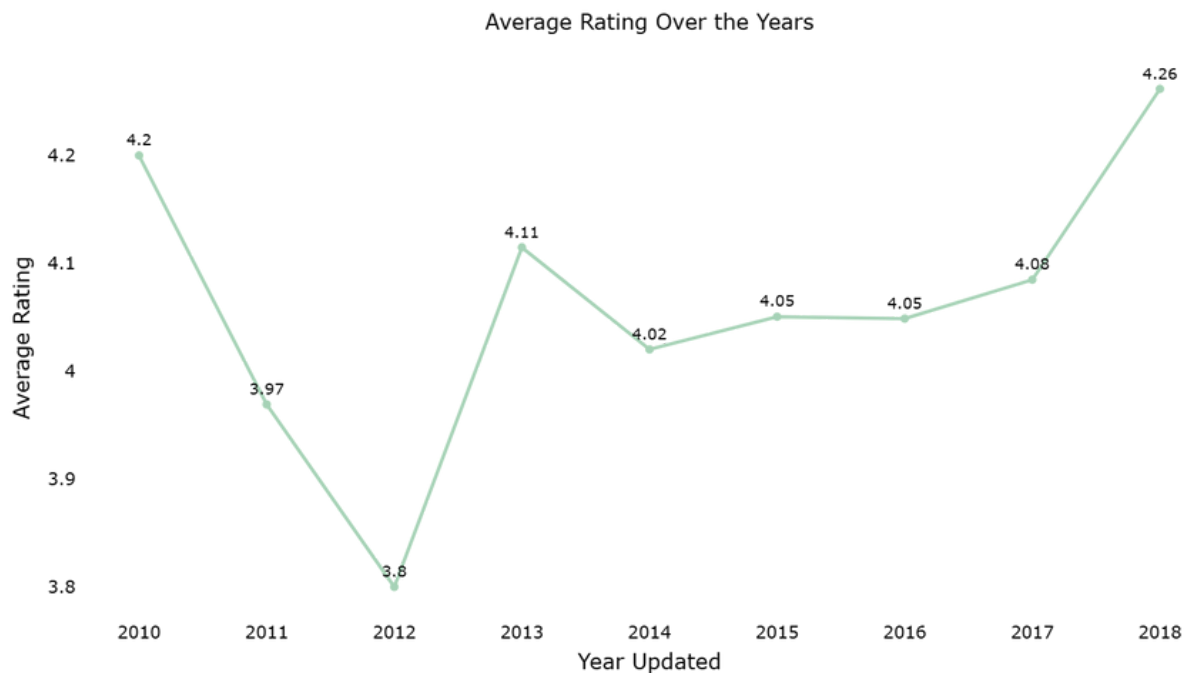
Key observations

- The Play Store ecosystem is heavily skewed toward free applications across all content ratings, with 'Everyone' dominating both free and paid types

Business Insights

- Paid apps constitute a small minority, indicating that upfront pricing is not the preferred monetization strategy regardless of content classification

Ratings Stabilize as Review Volume Increases



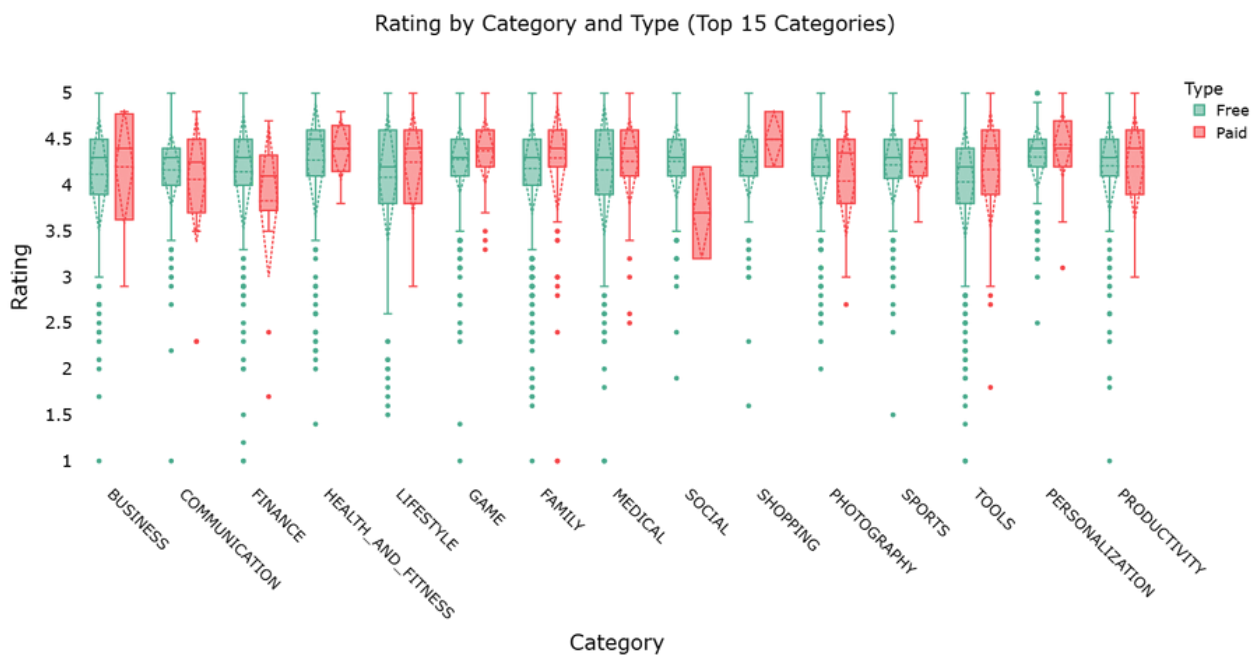
Key observations

- Average ratings stayed above 3.8 across all years, with a dip around 2011-2012 followed by recovery.
- From 2013 onwards, ratings show minimal fluctuation and a gradual upward trend, indicating stabilization.

Business Insights

- High and stable ratings suggest that the Play Store maintains consistent app quality and user satisfaction.
- The gradual upward trend signals platform maturity, indicating opportunities for user trust, app launches, and growth strategies.

Free Apps Dominate Most Categories, While Paid Apps Succeed Only in High-Trust or High-Value Segments



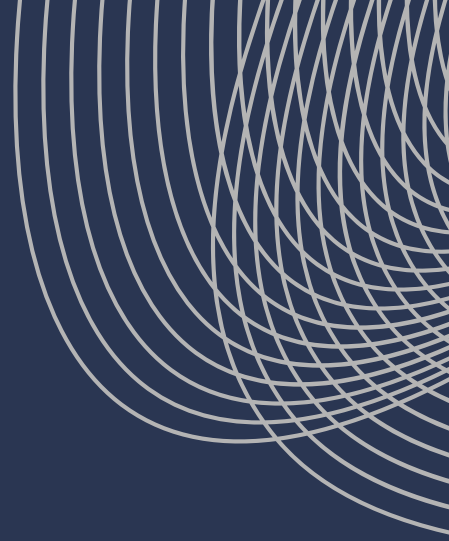
Key observations

- Free apps dominate most categories (Education, Art & Design, Events, House & Home) with high ratings and low variance, while Paid apps rarely show significant rating advantages.
- Paid apps outperform Free apps in certain sensitive or niche categories (Finance, Health & Fitness) and in some high-value categories (Food & Drink, Beauty), but Free apps remain essential for reach and virality (Games, Communication).

Business Insights

- Adopt Free-first or Freemium models in categories where Free apps are strong, leveraging optional premium features to reduce risk while maintaining user adoption.
- Use Paid or Premium-first models in high-trust or high-value categories to ensure quality perception, capitalize on user willingness to pay, and maintain credibility, while balancing risk and reach with hybrid models (Free to Paid upgrades) where appropriate.

BUSINESS / DEVELOPER TAKEAWAYS



- **High-Competition Categories:** Family, Games, and Entertainment dominate. Consider exploring less crowded categories like Auto & Vehicles, Events, and Beauty to stand out.
- **Free Apps Lead the Market:** Most apps are free. Freemium or ad-based models help reduce risk while maximizing reach.
- **Free vs. Paid Strategy by Category:**
 - **Free apps dominate most categories.** A free-first or freemium approach works best.
 - **Paid apps thrive in high-trust, high-value segments.** A paid- or premium-first strategy builds credibility and attracts willing buyers.
- **Target Broad Audiences:** “Everyone”-rated apps dominate. Focusing on general users helps boost downloads and ad revenue.



BUSINESS / DEVELOPER TAKEAWAYS

- **Optimize App Size & UX:** With a median size of ~14 MB and a mean of ~23 MB, balancing rich features with storage efficiency is key to avoiding download friction.
- **Ratings Require Volume & Quality:** Apps with few reviews show unreliable ratings. Early feedback helps stabilize perceived quality, while a consistent UX builds user trust.
- **Platform Maturity & Opportunity:** Ratings have stabilized over time. Leverage platform improvements to maintain quality, while exploring less crowded categories to increase visibility.



CHALLENGES & OPPORTUNITIES

Limitations

- The dataset is time bounded and may not reflect current trends or changes.
- Trending videos represent a small subset of all content, creating bias toward highly popular videos.
- Geographic coverage is limited to selected countries and may not capture global viewing behavior.
- Temporal relevance is constrained, as long-term performance and post-trending behavior are not captured.

Future Work / Opportunities:

- Perform sentiment analysis on video comments to understand audience perception and reactions.
- Conduct time-series analysis on views and likes to study how engagement evolves before and after trending.
- Compare trending videos with non-trending videos to identify factors driving trend eligibility.
- Analyze the impact of video duration, title keywords and thumbnails on engagement and reach.



CONCLUSION

This analysis of YouTube trending videos provided a comprehensive understanding of content trends, audience behaviour & performance characteristics. Key patterns and opportunities were identified, demonstrating how data driven insights can inform strategic decisions for content creators, marketers, and platforms. Future analysis could include sentiment analysis of viewer comments, time series tracking of engagement patterns, and comparisons with non trending or cross platform video data to uncover deeper insights.

This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.

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